

PRODUCTION ANALYSIS



Theory of Production

- **Production Function**

It refers to the physical relationship between the use of the inputs (factors of production) and resulting output of a product.

- It refers to the functional relationship, under the given technology, between physical rates of input and output of a firm, per unit of time.
- The relation between input and output of a firm has been called the production function. It includes all the technically efficient method of production.

Theory of Production

- The **production function** refers to the physical relationship between the inputs or resources of a firm and their output of goods and services at a given period of time, *ceteris paribus*.
- Physical output refers to the total amount of output produced in physical units (may refer to, kilograms of sugar, sacks of rice produced, etc).
- Inputs - are resources that contribute in the production of a commodity.
- Most resources are lumped into three categories:
 - Land,
 - Labor,
 - Capital.

The Production Function

- ***Production function***: the relationship that describes how inputs like capital and labour are transformed into output.
- Mathematically,

$$Q = F(K, L)$$

K = Capital, fixed factor.

L = Labour

Fixed vs. Variable Inputs

- Fixed inputs -resources used at a constant amount in the production of a commodity.
- Example: Capital Equipment, Machines and Land remain fixed in the short run. **Fixed inputs** cannot vary in the short run.
- Variable inputs - resources that can change in quantity depending on the level of output being produced. Example: Labour, raw materials are increased to expand output. **Variable input:** an input that can be varied in the short run.
- The longer planning the period, the distinction between fixed and variable inputs disappears, i.e., all inputs are variable in the long run.

Theory of production

- It describes the laws of production There are two laws of production
- 1. The law of variable proportions or law of diminishing returns (Short run analysis of production function):**

In the short run output may be increased by using more variable factors , while keeping fixed factors such as capital, land etc are held constant.
 - 2. Laws of returns to scale (Long Run analysis of production function).**
- Output can be increased by changing all the factors of production in the long run.

- **Total Product**

Total quantity of output produced by the given use of input in the production system, per unit of time.

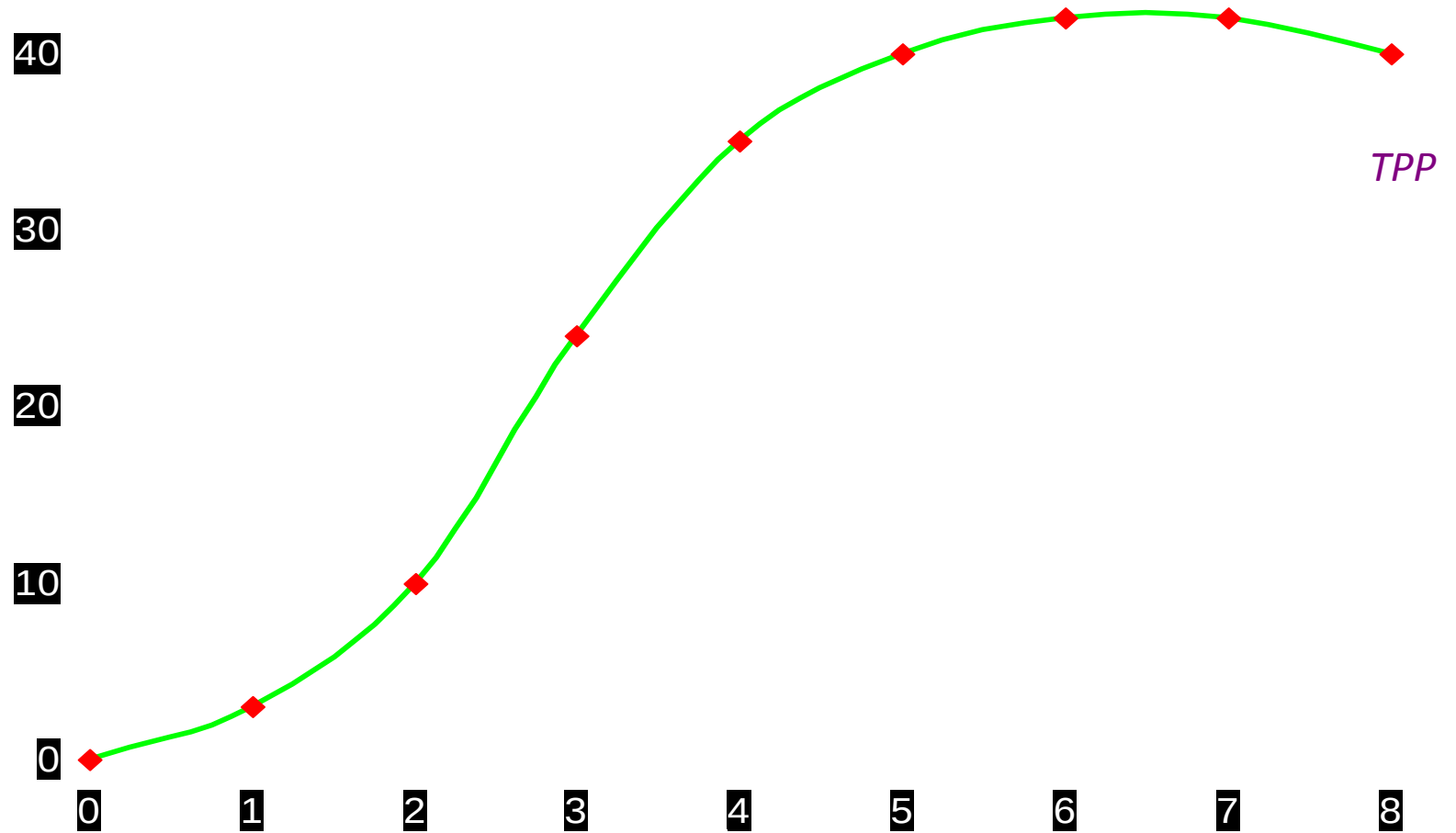
- **Marginal Product**

The addition made to the total product associated with a one-unit change in a particular single input labour.

- **Average Product**

Total quantity of product divided by the total units of input employed. It is per factor unit product.

TPP Curve



TPP

- TPP of a variable factor along with fixed factor continue to rise up to the point where the total output reaches at the maximum level.
- Further increase in employment causes to the total product to decline.
- Shift in the TPP: TPP curve is drawn on the assumption that the amount of fixed factor is held constant.
- Now any change in the fixed factor will cause a shift in the TPP curve of variable factor.
- APP: APP of a variable factor is the total TPP divided by the amount of labour employed with a given quantity of fixed capital used to produce a commodity

Law of Diminishing Returns (Diminishing Marginal Product)

The law of diminishing returns states that when more and more units of a variable input are applied to a given quantity of fixed inputs, the total output may initially increase at an increasing rate and then it will eventually increase at diminishing rates.

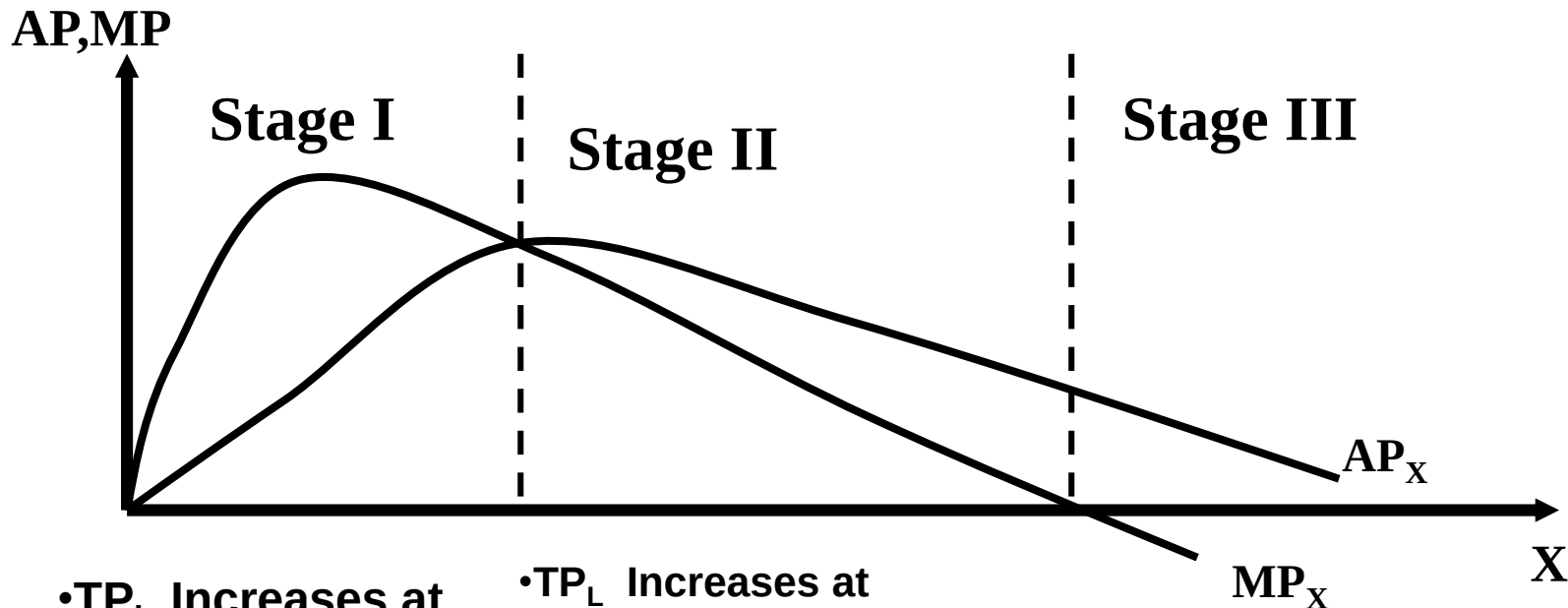
Assumptions. The law of diminishing returns is based on the following assumptions: (i) the state of technology is given (ii) labour is homogenous and (iii) input prices are given.

Law of Variable Proportion or Diminishing Marginal Return

- **Statement.** Using the concept of marginal product, the law may be stated as follows:

when the units of a variable factor are increased along with fixed factors, the total product initially may rise at an increasing rate and, after a point, it tends to increase at a decreasing rate because the marginal product of the variable factor in the beginning may tend to rise but eventually tends to diminish.

Three Stages of Production in Short Run



- TP_L Increases at increasing rate.

- MP Increases at decreasing rate.

- AP is increasing and reaches its maximum at the end of stage I

- TP_L Increases at Diminishing rate.

- MP_L Begins to decline.

- TP reaches maximum level at the end of stage II, $MP = 0$.

- AP_L declines

- TPL begins to decline

- MP becomes negative

- AP continues to decline

Application of Law of Diminishing Returns:

- It helps in identifying the rational and irrational stages of operations.
- It gives answers to question –
How much to produce?
What number of workers to apply to a given fixed inputs so that the output is maximum?